



QUICKBITE

EXPRESS

Food Business Data Analytics Project

Data Analytics & Business Intelligence



Python



Pandas



NumPy



Matplotlib



Seaborn



Jupyter
Notebook



MySQL

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PROJECT OVERVIEW

QuickBite Express is a **Data Analytics** and **Business Intelligence** project focused on analyzing food-business data.

The project covers:



Customers



Restaurants



Menu Items



Orders



Order Items



Deliveries



Delivery Partners



Customer Reviews

MAIN GOAL



Transform business data into meaningful **insights** and **decision support**.



KEY POINT

The project uses **Python** and **MySQL** as complementary analytical tools to generate reliable business insights.



BUSINESS PROBLEM

Food-delivery businesses generate large volumes of operational and customer data.

Without structured analysis, it is difficult to identify:

-  Customer ordering patterns
-  High-demand menu items
-  Restaurant performance
-  Order activity
-  Delivery performance
-  Customer feedback
-  Data-quality issues

BUSINESS NEED



A structured analytical approach is required to convert raw data into reliable business information.



PROJECT OBJECTIVES

The main objectives of this project are to analyze food-business data using Python and SQL to generate **meaningful insights** for business decision-making.

1



Clean and prepare business datasets.

Perform data cleaning, handling missing values, duplicates and inconsistencies.

2



Perform data-quality checks.

Validate accuracy, consistency, data types and business rules.

3



Conduct Python-based EDA.

Explore data distributions, patterns, relationships and key trends.

4



Create meaningful visualizations.

Use charts and graphs to represent insights clearly and effectively.

5



Prepare cleaned data for SQL analysis.

Import cleaned datasets into MySQL for structured querying.

6



Perform structured MySQL analysis.

Write and execute SQL queries to answer business questions and generate KPIs.

7



Validate important business conditions.

Check data integrity, financial accuracy and operational consistency.

8



Generate business insights.

Derive actionable insights from analysis and uncover key opportunities.

9



Develop practical recommendations.

Provide data-driven recommendations to support business growth and efficiency.

10



Interpret findings as business insights.

Convert analytical results into meaningful information for decision support.

11



Document the complete analytical workflow.

Maintain reproducible process, queries, observations and project documentation.



OUR GOAL



Transform data into **actionable insights** and better business decisions.



KEY TAKEAWAY

These objectives guide the end-to-end analytical journey from raw data to reliable insights that **drive business value**.

The project uses eight Python-cleaned datasets that cover customers, restaurants, menu items, orders, deliveries and reviews.



EIGHT CLEANED DATASETS

#	DATASET	RECORDS	BUSINESS AREA	DESCRIPTION
1	 customers_clean	107,776	Customers	Customer profiles and acquisition information
2	 restaurants_clean	19,995	Restaurants	Restaurant details and operational information
3	 menu_items_clean	342,671	Menu Items	Menu items, categories, prices and veg/non-veg
4	 delivery_partners_clean	15,000	Delivery Partners	Delivery partner profiles and vehicle information
5	 orders_clean	149,166	Orders	Order details, amounts and status information
6	 order_items_clean	342,994	Order Items	Items in each order with quantity and pricing
7	 delivery_clean	149,166	Delivery	Delivery performance and SLA information
8	 reviews_clean	68,825	Reviews	Customer reviews, ratings and sentiment



TOTAL RECORDS ACROSS ALL DATASETS: **1,195,593**

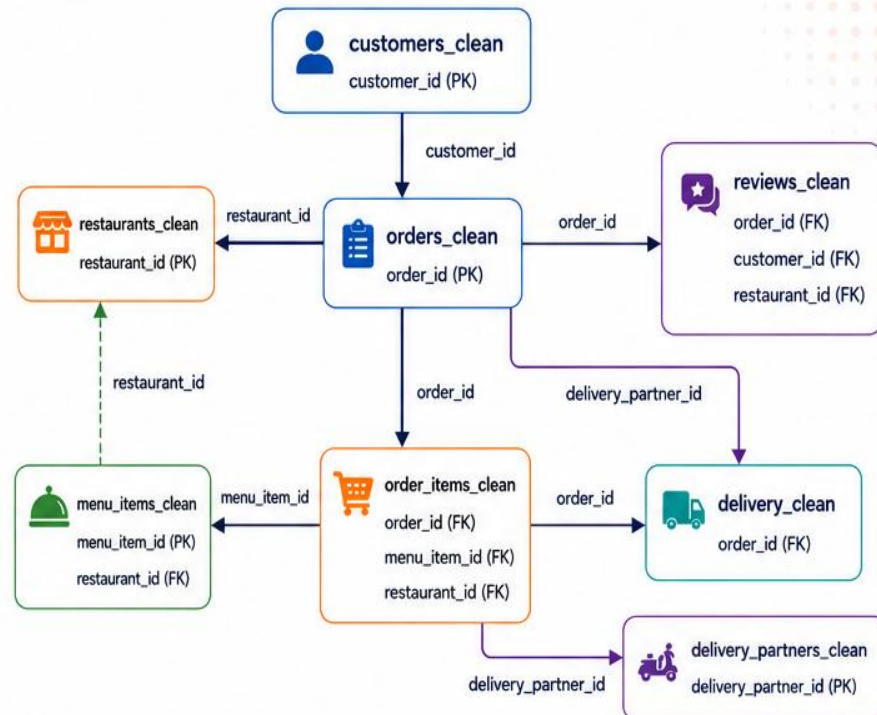


KEY TAKEAWAY

These eight datasets are **connected through key relationships**, forming a structured database that enables **comprehensive analysis** of the QuickBite Express food-delivery business.



DATASET RELATIONSHIP DIAGRAM



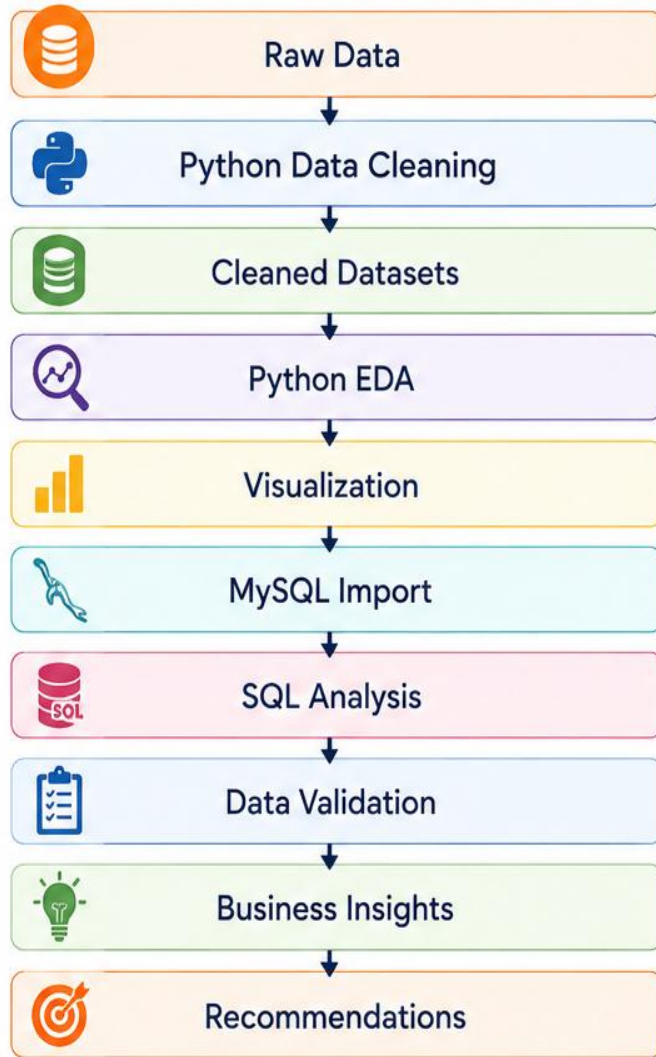
RELATIONSHIP KEYS

PK : Primary Key

FK : Foreign Key

→ : Relationship

PROJECT WORKFLOW



KEY POINT



Python and SQL were used as **complementary** analytical tools.

DATA CLEANING & PREPARATION



PYTHON DATA PREPARATION

Python and Pandas were used for:



Dataset inspection



Missing-value checks



Duplicate checks



Data-type validation



Categorical consistency



Numerical validation



Data transformation



Financial reconciliation



Saving cleaned datasets



FINANCIAL VALIDATION



orders_clean includes:

total_amount



calculated_total



total_difference



order_items_clean includes:

line_total



calculated_line_total



line_total_difference

OUTCOME



Cleaned datasets became the **controlled input** for further analysis.

EXPLORATORY DATA ANALYSIS



PYTHON EDA

EDA was performed to understand:



Customer activity



Restaurant activity



Menu-item activity



Order patterns



Delivery activity



Delivery-partner information



Review patterns



Numerical distributions



Categorical frequencies



PURPOSE

Identify **important patterns, differences,** and **areas requiring further investigation.**



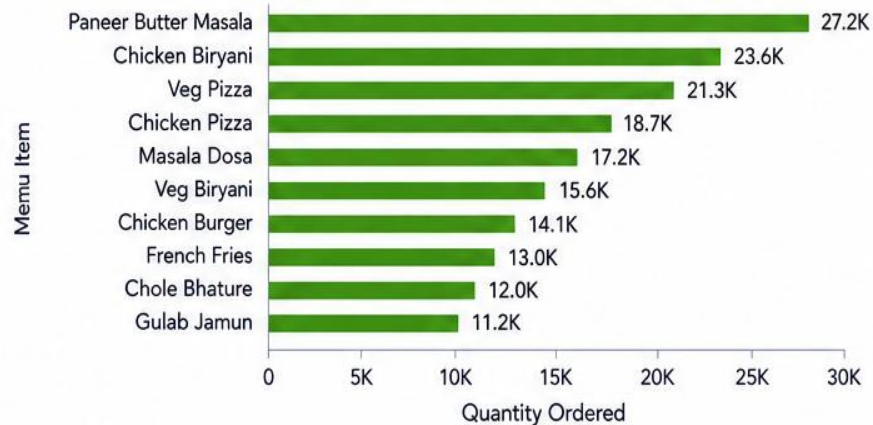
SAMPLE EDA INSIGHTS

Monthly Order Trend



Order volume increases from Jan to Jul and then gradually declines.

Top 10 Menu Items by Quantity Ordered



A small group of menu items contributes to a large share of ordered quantity.

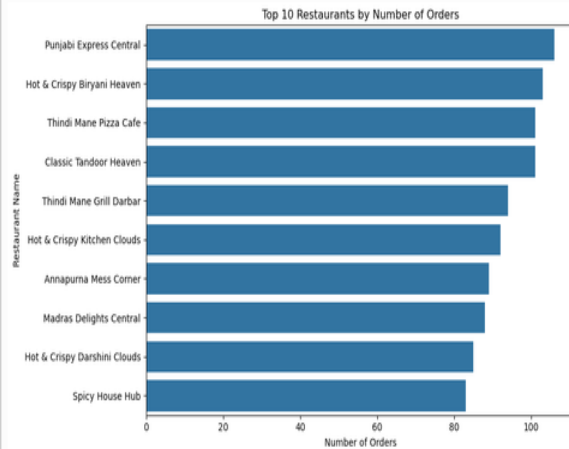


KEY TAKEAWAY: Clear differences exist in order volume over months and in demand across menu items.

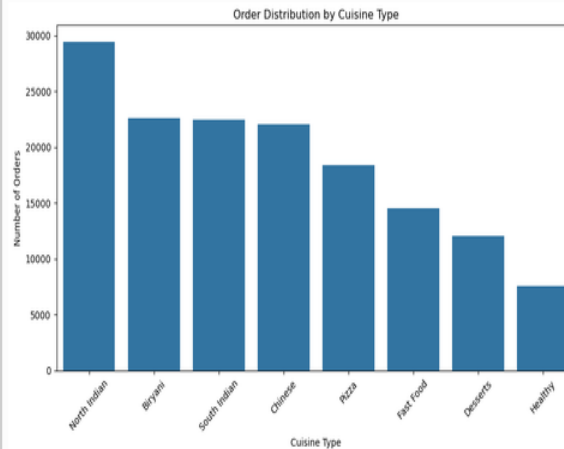
DATA VISUALIZATION — PART 1

Key demand and product-performance visualizations — original charts from the IPYNB

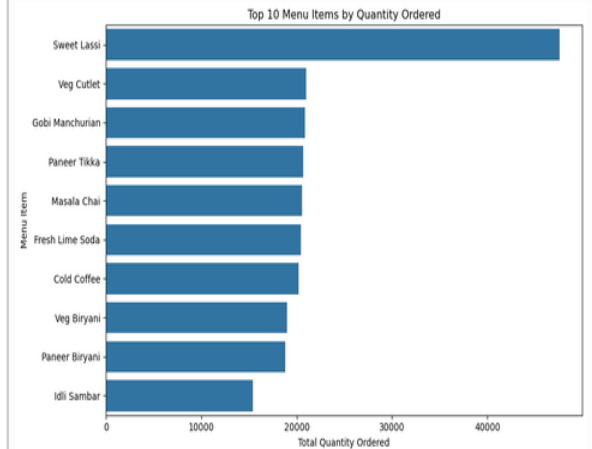
RESTAURANT DEMAND



CUISINE DEMAND



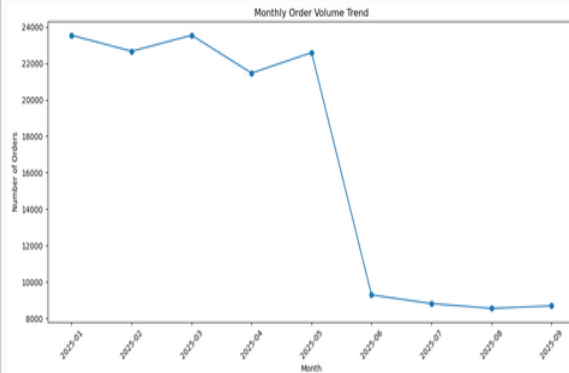
MENU ITEM DEMAND



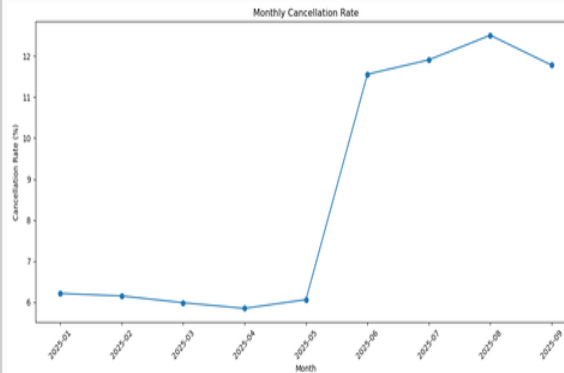
DATA VISUALIZATION — PART 2

Key trend, operational and customer-experience visualizations — original charts from the IPYNB

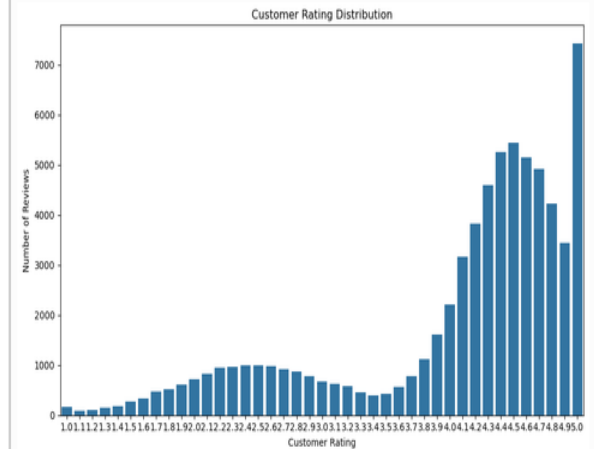
MONTHLY ORDER TREND



CANCELLATION RATE



CUSTOMER RATINGS



SQL DATABASE & BUSINESS ANALYSIS

MYSQL DATABASE

Database: **quickbite_express**

8 cleaned tables imported from Python:

- customers_clean
- restaurants_clean
- menu_items_clean
- delivery_partners_clean
- orders_clean
- order_items_clean
- delivery_clean
- reviews_clean

Python-cleaned CSVs → MySQL import → SQL analysis

SQL TECHNIQUES USED

- SELECT • WHERE • ORDER BY
- GROUP BY • HAVING • DISTINCT
- Aggregate functions • CASE
- INNER JOIN • LEFT JOIN
- Subqueries • CTEs
- ROW_NUMBER() • RANK()
- PARTITION BY • Window functions
- Conditional aggregation

MAIN BUSINESS ANALYSIS AREAS

- Customer & restaurant performance
- Menu & food-item demand
- Orders, revenue & cancellation
- Delivery & SLA performance
- Reviews & customer ratings
- Integrated cross-table analysis

KEY POINT: SQL converts the cleaned Python datasets into structured, business-oriented analysis.

DELIVERY PERFORMANCE

ACTUAL DELIVERY DATA

delivery_clean fields used:

actual_delivery_time_mins

Actual delivery duration

expected_delivery_time_mins

Expected delivery duration

delivery_delay_mins

Delivery delay

distance_km

Delivery distance

sla_met

SLA status

SQL ANALYSIS FOCUS

1

SLA Performance

Within SLA vs Outside SLA

2

Average Delivery Time

Overall delivery-time baseline

3

Fastest Deliveries

Lowest actual delivery times

4

Longest Deliveries

Highest actual delivery times

5

City-wise Performance

Average time and delay by city

6

Partner Performance

Workload, time, delay and rating

KEY INSIGHT AREA

Delivery analysis connects time, delay, distance, SLA and partner performance for operational decisions.

CUSTOMER & BUSINESS INSIGHTS

Evidence-based interpretation from Python visualization and SQL analysis

CUSTOMER ACTIVITY

Customer-order analysis identifies differences in ordering frequency and activity.

Business use: retention and customer engagement.

MENU / PRODUCT DEMAND

Menu-item analysis identifies higher- and lower-activity items.

Business use: product availability and inventory planning.

RESTAURANT PERFORMANCE

Restaurant-level order analysis compares activity across restaurants.

Business use: targeted operational attention.

ORDER & DEMAND PATTERNS

Order analysis reveals overall transaction activity and ordering patterns.

Business use: demand and performance monitoring.

DELIVERY & CUSTOMER FEEDBACK

Delivery fields support time, delay and SLA analysis; reviews

IMPORTANT: Insights are presented without inventing numerical results; exact values remain in the notebook/SQL outputs.

DATA VALIDATION & QUALITY

Checks performed across Python preparation and SQL verification

MISSING VALUES

Checked important fields during Python preparation and, where required, after SQL import.

Purpose: identify incomplete records that may affect calculations.

DUPLICATE RECORDS

Duplicate records were checked during Python cleaning.

Purpose: prevent inflated counts and totals.

DATA TYPES

Identifiers, dates, quantities, prices, ratings and other analytical fields were reviewed.

Purpose: support reliable calculations, filtering and grouping.

CATEGORICAL & NUMERICAL CHECKS

Distinct-value checks and numerical-range checks were used to identify unexpected values.

Purpose: detect consistency or business-rule issues.

SQL IMPORT VALIDATION

Record counts, sample rows, column names, data types and required fields were verified.

Purpose: confirm the cleaned data was imported correctly.

FINANCIAL RECONCILIATION

Orders: total_amount ↔ calculated_total ↔ total_difference. Order items: line_total ↔ calculated_line_total ↔ line_total_difference.

Purpose: assess financial consistency without silently changing original financial values.

VALIDATION OUTCOME: Improved confidence in the reliability and completeness of the analytical results.

BUSINESS RECOMMENDATIONS

Actionable decisions derived from Python EDA and SQL business analysis



1. PRIORITIZE HIGH-REVENUE MARKETS

- Focus marketing, restaurant partnerships and delivery capacity on cities generating higher completed-order revenue.
- Allocate more resources to high-performing cities for growth.



2. STRENGTHEN CUSTOMER RETENTION

- Target high-value and repeat customers with loyalty rewards, personalized offers and retention campaigns.
- Monitor repeat-customer rate as a key performance indicator.



3. PROMOTE HIGH-DEMAND FOOD ITEMS

- Give popular menu items greater visibility through featured listings and combo offers.
- Run category-based promotions to drive demand for top-selling items.



4. DEVELOP STRATEGIC RESTAURANT PARTNERSHIPS

- Prioritize restaurants with strong order volume, revenue and customer ratings.
- Review high-revenue but lower-rated restaurants for service and quality improvement.



5. IMPROVE DELIVERY OPERATIONS

- Review slower locations and delivery delays to remove operational bottlenecks.
- Allocate more orders to high-performing delivery partners and balance partner workload.



6. MONITOR CUSTOMER SATISFACTION & CANCELLATION

- Use ratings, reviews and sentiment to identify restaurants and areas requiring improvement.
- Analyze cancellation patterns by city, restaurant and category and take corrective actions.



RECOMMENDATION OUTCOME:

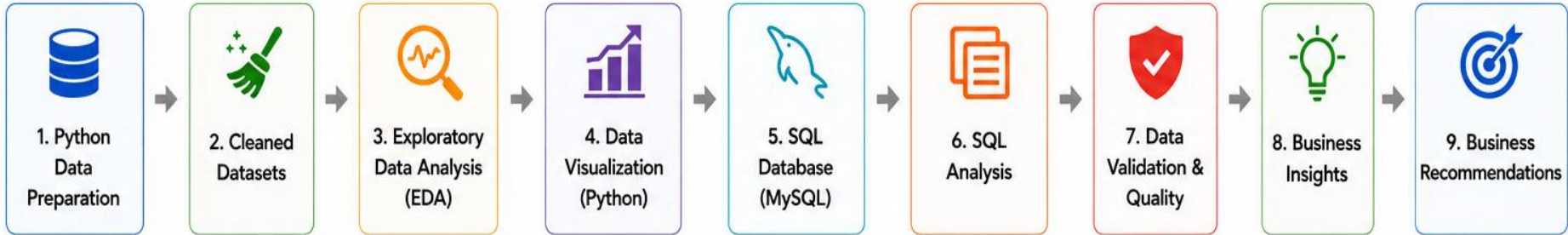
Turn analytical findings into targeted actions for growth, retention, operational efficiency and customer experience.



CONCLUSION

QuickBite Express demonstrates a complete Data Analytics and Business Intelligence workflow using Python and MySQL.

OUR ANALYTICAL JOURNEY



PROJECT OUTCOMES

- ✓ Cleaned and validated data across 8 business datasets.
- ✓ Comprehensive EDA and **12** meaningful visualizations revealing important patterns.
- ✓ Structured SQL analysis using joins, aggregations, CTEs, window functions and validation queries.
- ✓ Business insights on customers, restaurants, menu items, orders, deliveries, delivery partners and reviews.
- ✓ Evidence-based recommendations for customer retention, product planning, restaurant operations and delivery efficiency.

BUSINESS VALUE DELIVERED



CUSTOMER INTELLIGENCE

Understand customer activity, high-value segments and repeat customers.



RESTAURANT PERFORMANCE

Identify high-performing and low-rated restaurants for strategic actions.



FOOD & MENU INSIGHTS

Highlight popular items and category-wise demand patterns.



DELIVERY OPERATIONS

Analyze delivery times, delays, SLA performance and partner efficiency.



CUSTOMER EXPERIENCE

Understand ratings, reviews and sentiment for quality improvement.



FINAL OUTCOME

The project demonstrates how structured business data can support data-driven decision making across all key areas of the food business.



BUSINESS IMPACT

Data-driven decision support across customer, product, restaurant, order, delivery and review operations.

Thank You!

Questions & Discussion

